ARCHITECTURE & UI BLUEPRINT

Project Lumen

An Open-Source, Hardware-Accelerated RAW Editor for macOS & Apple Silicon

TARGET ARCHITECTURE	FRAMEWORK FOCUS	DOCUMENT VERSION
Apple Silicon (M1/M2/M3/M4)	SwiftUI, Metal, CoreML	v1.0 (Comprehensive)

1. Design Language & Interface System

Project Lumen should feel entirely integrated into macOS, embracing a professional, tactile dark interface that steps away from generic web-like components. By building on native `AppKit` and `SwiftUI` hierarchies, the interface maintains precise sub-pixel rendering, low-latency layout loops, and perfect hardware integration.

1.1 Interface Hierarchy & Canvas Optimization

- **Unified Window Model:** Utilize a single-window interface with sliding sidebars (using native `NavigationSplitView`) to prevent window fragmentation. Sidebars should collapse cleanly using macOS shortcuts (`⌘+B` for Library, `⌘+D` for Develop).
- **Color Spectrum Control:** The layout must default to a dark interface variant tailored specifically for professional color matching. Use a calibrated neutral dark gray background (`#161616`) to prevent optical contrast illusions, keeping text at soft ivory (`#e2e8f0`) to reduce eye fatigue over multi-hour editing sessions.
- **Metal Canvas Layering:** The photo viewport must skip standard drawing cycles. Embed a custom `CAMetalLayer` within an `NSViewRepresentable` container. This forces the image tile renderer to communicate straight to the Apple Silicon GPU framework, eliminating interface layout delay.

> **TACTILE HARDWARE INTERACTION BLUEPRINT**
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> **Dial & Slider Kinematics:** When tracking trackpad gestures or fine cursor movements across adjustments (like Exposure or Clarity), map adjustments to logarithmic scales. A small initial shift modifies values by tight steps (e.g., 0.01 eV), while rapid acceleration scales up linearly, replicating the tactile dampening of physical luxury dials.

2. Deep-Dive Metal Compute Pipeline

To outpace legacy cross-platform apps, Project Lumen must route image channels completely inside the Apple Silicon Unified Memory System. Traditional architectures cycle image data back and forth from the CPU cache to independent VRAM spaces. By writing custom Metal Compute Shaders (`.metal` files), memory allocations are bound exactly once.

2.1 Core Demosaicing Pipeline

When reading a raw frame, the application skips high-overhead interpreter packages and decodes pixel channels directly into single planes. A bilinear or directional high-pass Metal compute kernel performs hardware-accelerated demosaicing.

```
// Conceptual Metal Kernel for Demosaicing
#include <metal_stdlib>
using namespace metal;

kernel void demosaicBayer(texture2d<float, access::read> rawInput [[texture(0)]],
                          texture2d<float, access::write> rgbOutput [[texture(1)]],
                          uint2 gid [[thread_position_in_grid]]) {
    // Spatial interpolation mapping color filters across the grid matrix
    float4 centerSample = rawInput.read(gid);
    // Interpolate neighboring channels using low-latency local execution cache...
```

```
    rgbOutput.write(centerSample, gid);
}
```

2.2 The Global Slider Transformation Engine

Every active adjustment panel updates an instruction struct array passed to the GPU execution context. This allows changes to calculate on the fly across multiple exposures instantly.

Adjustment Engine	Underlying Shader Operation	Mathematical / Linear Matrix Formula
Exposure	Linear Channel Multiplication	*C_{out} = C_{in} imes 2^{\Delta eV}*
Contrast	Sigmoidal Midtone S-Curve Mapping	*C_{out} = rac{1}{1 + e^{- lpha(C_{in} - 0.5)}}*
3-Way Grading	Luminance-Weighted Lift/Gamma/Gain	*C_{out} = (C_{in} imes G_{gain} + S_{shadows}) imes (1 - W_{mid})*

3. The Smart Library Architecture (SQLite + CoreData Canvas)

A photographer's sorting speed hinges entirely on sub-millisecond data indexing. Lumen integrates an absolute dual-tier storage strategy designed specifically for internal and external storage arrays.

3.1 Memory Mapped Caching & Pre-renders

- **Tier 1 — Thumbnail Cache:** Store pre-rendered 8-bit Display-P3 previews inside a unified memory-mapped file database. This enables thousands of miniature images to render instantaneously onto layout tracks when moving sliders or scrolling fast.

  Tier 2 — The Managed Database: Manage organizational metadata utilizing a hyper-optimized SQLite layout compilation. Explicit index assignments across critical criteria (such as ISO speed, Camera Make, Lens type, Rating tags, and Flag values) bring catalog searches down to fractional millisecond runtimes.

3.2 Keyboard-Driven Layout Action System

To maintain efficient work streams, design clear mechanics for one-handed operation models. Left hand manipulates organizational groupings, right hand pilots image selection sequences.

- `1` – `5`: Apply Star values directly across selected photo assets.
- `6` – `9`: Assign localized organizational color tabs to group categories.
- `P` / `X`: Tag assets instantly with Select (Pick) or Reject flags. Reject flags dim the workspace target thumbnail by an adjustable opacity layer (e.g., 40%).

4. Advanced Local Masking & CoreML Engine

Spatial adjustments utilize deep on-device CoreML neural patterns, processing structures seamlessly by executing directly inside the integrated Apple Neural Engine (ANE).

4.1 Neural Foreground Separation & Sky Masking

Instead of generic boundary routines, load an ultra-lightweight, quantized instance of a semantic segmentation model (like an optimized MobileNetV4 or custom U-Net variant) converted straight into a `.mlmodelc` target compilation package.

- **The Sky Isolator:** Returns a crisp, monochrome alpha layout map of sky regions in less than 45ms on an M-series base platform. It applies automated linear edge smoothing to shield complex features like power lines or fine tree leaves.
- **Subject Extraction Matrix:** Isolates central human elements or defined focal foreground subjects, caching the generated bitmap directly alongside metadata properties to ensure real-time local tuning operations.

4.2 Precision Non-Destructive Local Brushes

Local adjustments avoid baking modifications into persistent graphics textures. Instead, the brush builds a vector paint stroke layer graph. Each brush trace keeps track of relative viewport scaling factors, geometric path maps, and pressure values. The composite mask maps dynamically onto the active layout context using a specialized Metal overlay pass.

5. Advanced Tool Innovations for Apple Silicon

To set Project Lumen apart from legacy applications, the engine can implement three unique, deeply specialized tools built exclusively for high-bandwidth unified memory architectures.

5.1 Real-Time "Focus Plane" Heatmapping

While culling hundreds of sports or portrait shots, opening images individually to check focus is a major bottleneck. This tool runs a parallel Metal high-pass convolution filter across all image grid items simultaneously. It isolates sharp contrast transitions and renders a real-time, translucent green micro-mesh overlay directly over the in-focus areas inside the thumbnail view. Photographers can judge tack-sharp focus instantly without leaving the grid.

5.2 Micro-Texture Separation Engine (Frequency Separation Mixer)

Rather than basic slider operations, this tool maps high-resolution image data into dual frequency bands directly on the GPU:

- **Low Frequency Engine:** Isolates tonal transitions and pure chromatic data blocks.
- **High Frequency Engine:** Isolates structural contours, pores, fine hairs, and fabric weaves using localized Laplacian pyramids.

Photographers can smooth skin blemishes or reduce lens artifacts by adjusting the high-frequency contrast threshold without modifying the underlying, natural skin-tone gradients.

5.3 Predictive CoreML Smart Culling Assistant

An integrated ANE routine that analyzes raw image queues upon ingestion. The engine assigns an automated "Clarity & Composition Score" by calculating mathematical focus thresholds, tracking eye-blink states on human subjects, and monitoring exposure distributions. It automatically stacks near-identical burst shots together, pre-selecting the sharpest frame as the stack champion to drastically reduce culling workloads.

6. High-Throughput Export Pipeline

The export execution stack converts parametric adjustment instructions into a continuous processing matrix. By taking advantage of dedicated Apple Silicon hardware ProRes, HEVC, and JPEG processing engines, export pipelines run entirely in parallel without freezing user interface loops.

- **Grand Central Dispatch (GCD) Queuing:** Distributes rendering threads across specialized Performance (P) and Efficiency (E) processor cores. Heaviest calculations route exclusively to the Metal engine, keeping E-cores completely free to handle UI interactions and catalogue indexing operations.
- **Adaptive Sharpness Calibration:** Computes output-specific high-pass sharpening parameters tailored automatically to target print dimensions, high-resolution display standards, or highly-compressed web formats.